# Reason. Plan. Act. Physical AI Workshop

## Lecture 2b: Day 2 Conclusion

Sarah Keren

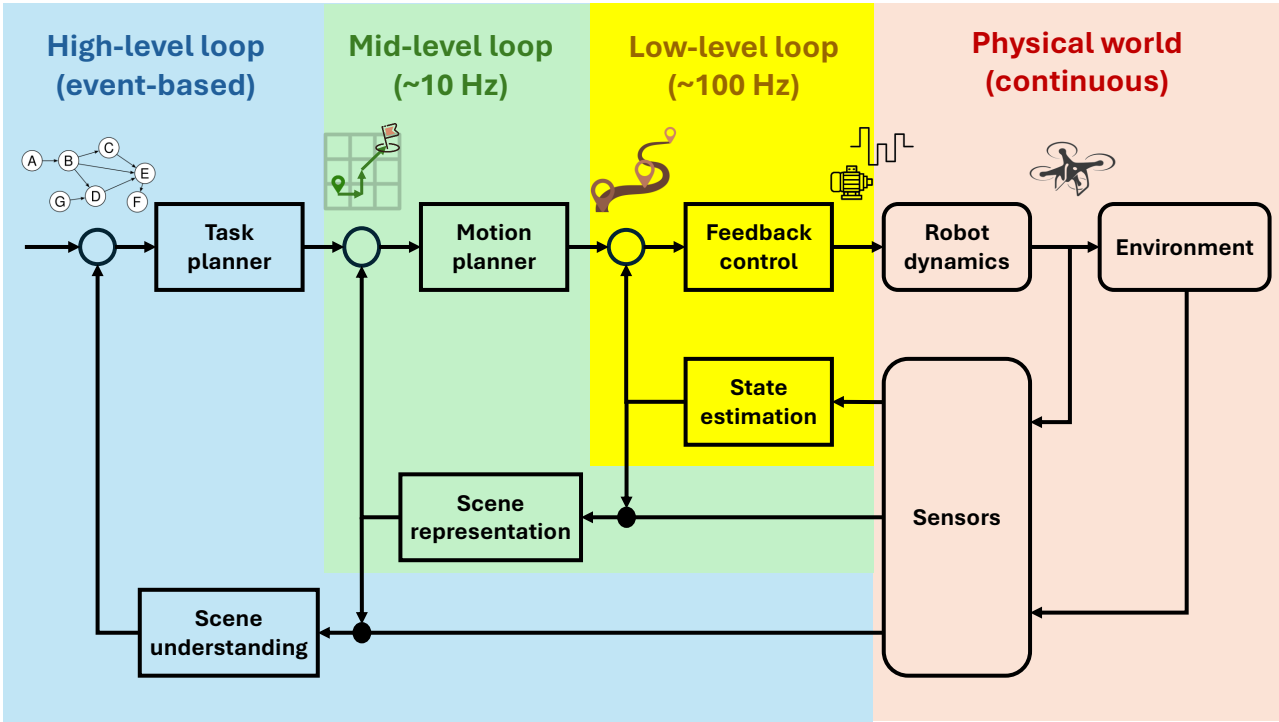
The Taub Faculty of Computer Science
Technion - Israel Institute of Technology



# Recap

# Layered Decision-Making





# Perception and Scene Understanding





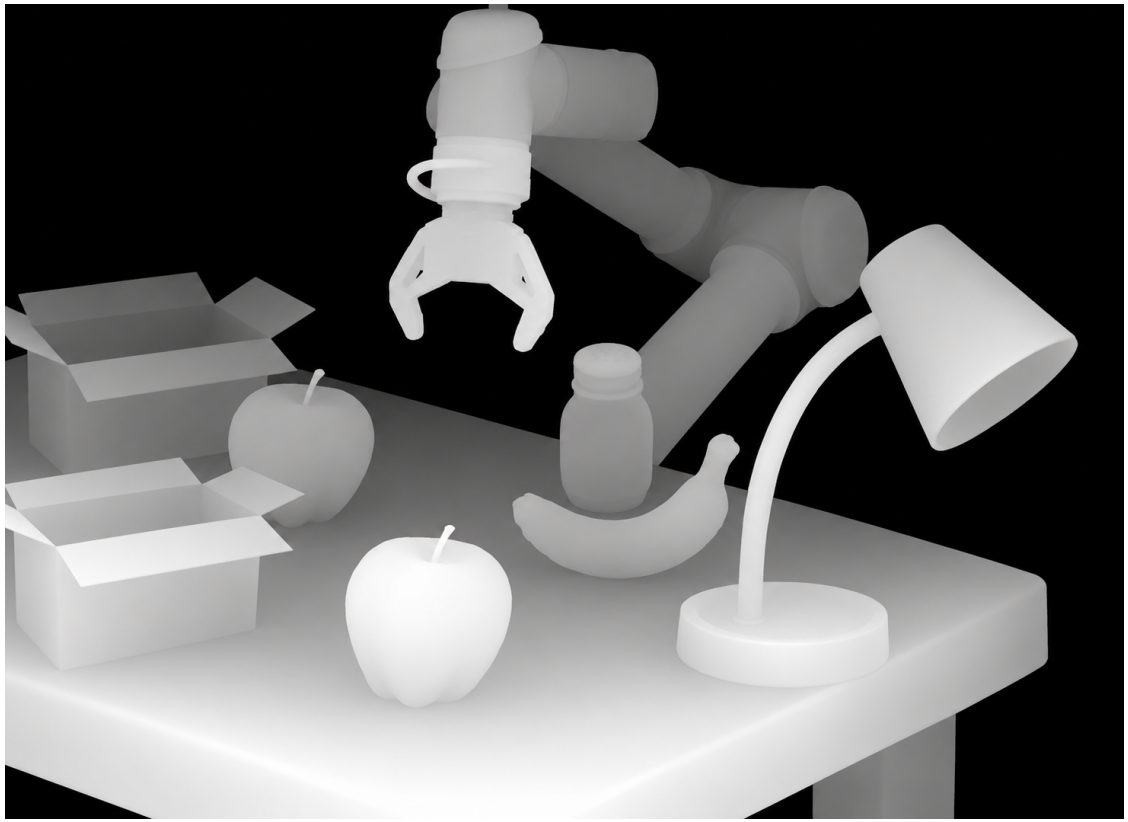
# Perception and Scene Understanding



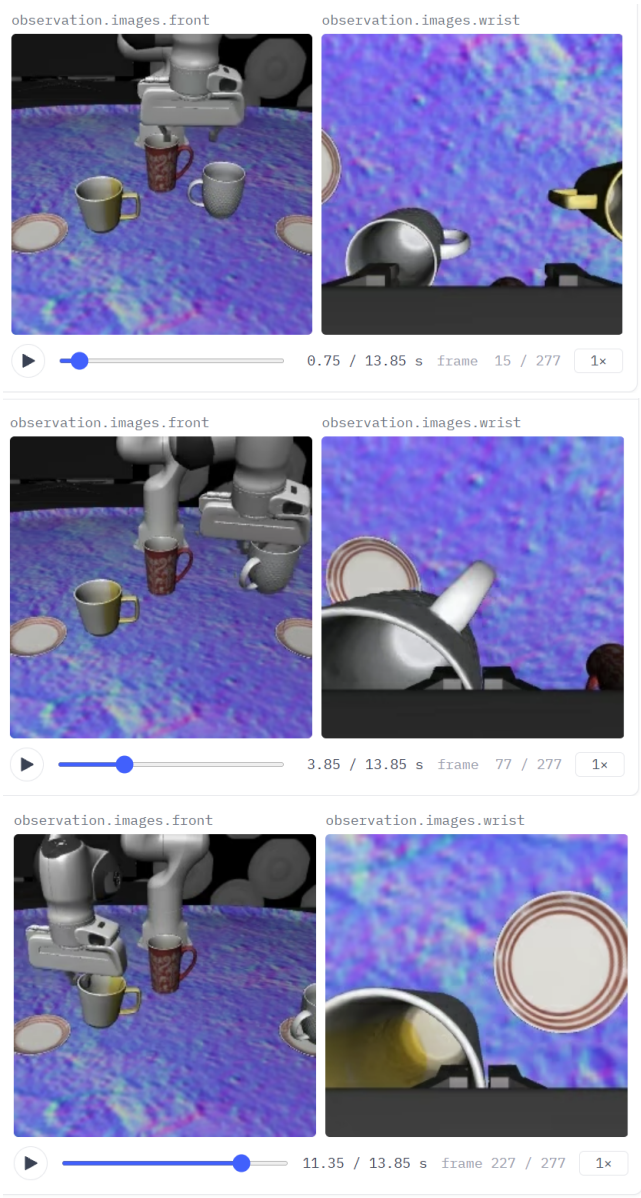


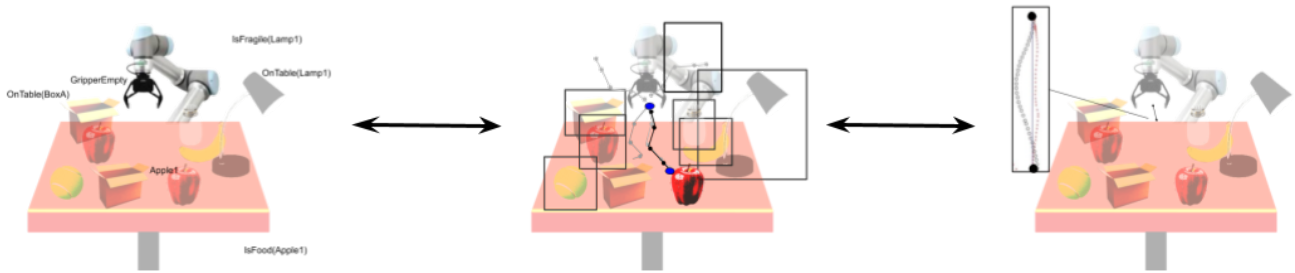
# Perception and Scene Understanding





# Perception and Scene Understanding

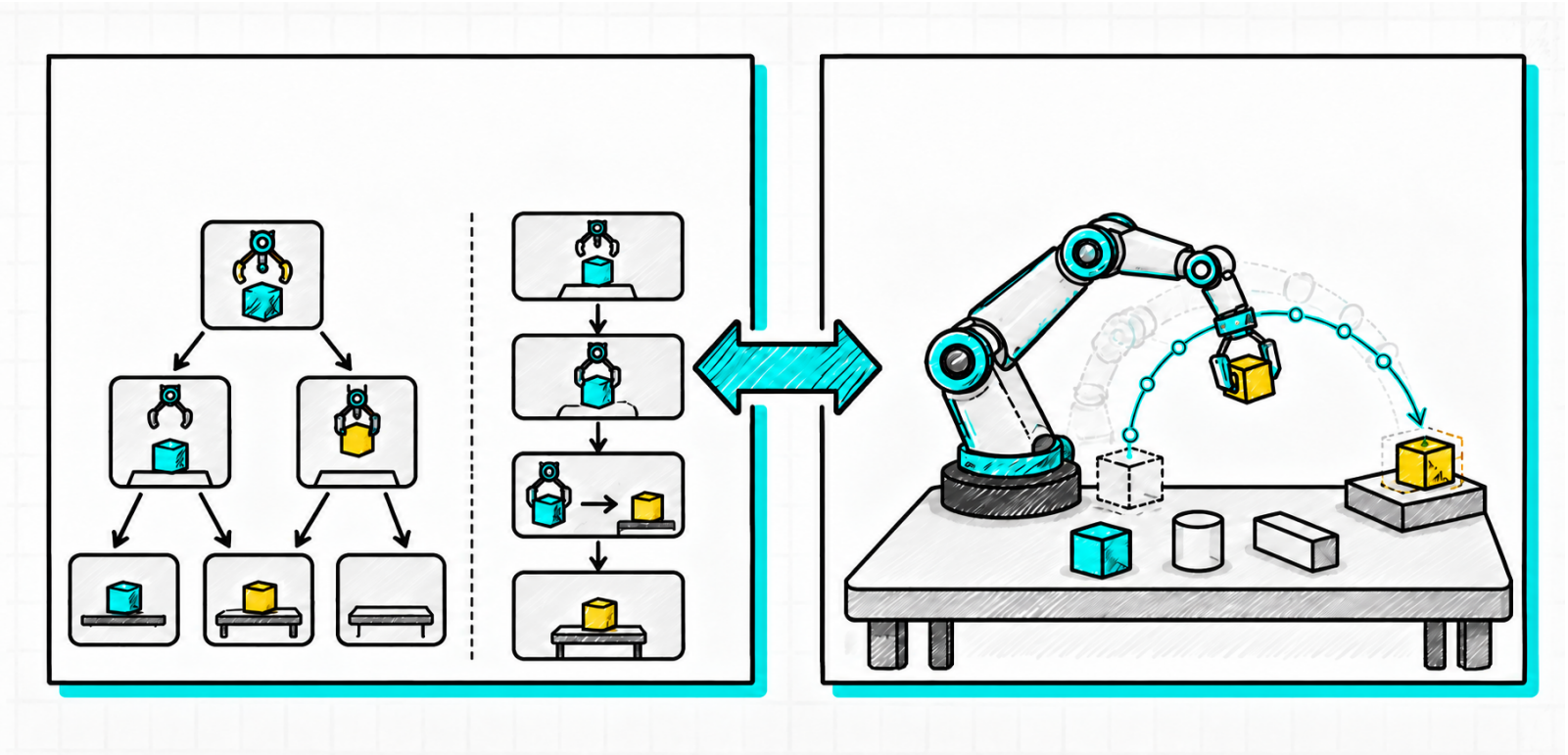



IsFragile(Lamp1)
GripperEmpty
OnTable(Lamp1)
OnTable(BoxA)
Apple1
IsFood(Apple1)

# Why Task Planning Matters in Robotics



**The robotics challenge:** A logically valid plan must also be physically feasible.



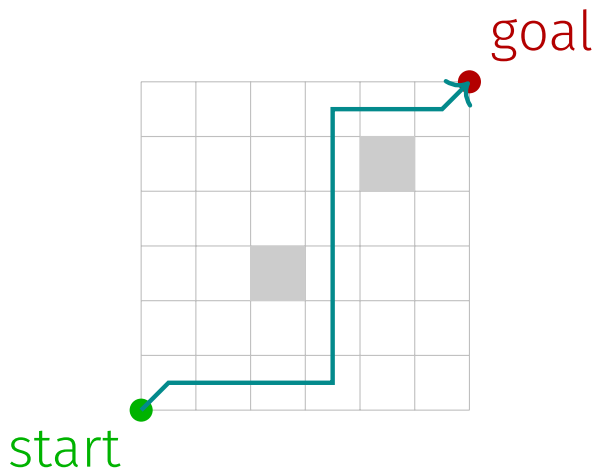
# Search-Based Motion Planning



**Idea:** Discretize $\mathcal{C}_{free}$ into a grid and search for a path.

Typical methods:

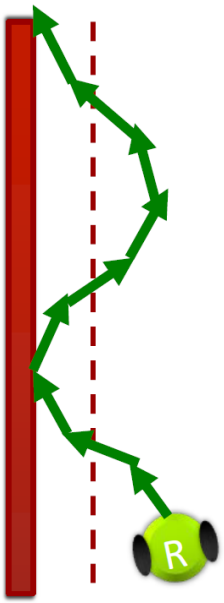
- **Breadth-first search (BFS)** — shortest path on uniform grid.
- **A* search** — uses heuristic $h(q)$ to guide exploration.




Advantages: simple, optimal on grid.
Limitations: exponential growth with dimension.

# Proportional (P) control






**Proportional Control Program Loop**

```
Move 0.5 body-length forward
If distance-to-wall > desired,
    let error = |desired – distance-to-wall|
    Then turn Kp * error towards the wall
    Else turn  Kp * error away from the wall
```

# Challenges



# Challenges in Task Planning



Find a sequence of actions that achieves a high-level goal.

- **Combinatorial explosion:** the number of possible action sequences grows exponentially with plan length.
- **Interacting subgoals:** achieving one condition may undo another or block future actions.
- **Modeling the world:** specifying relevant predicates, action preconditions, effects and costs requires careful abstraction.
- **Physical feasibility:** a symbolically valid action may have no feasible motion.
- **Uncertainty and change:** incomplete knowledge and unexpected outcomes require monitoring and replanning.

# Challenges in Task Planning



**Physical feasibility:** a symbolically valid action may have no feasible motion.



What do costs represent?

# Challenges in Motion Planning



Find a feasible path through a continuous configuration space.

- **High dimensionality:** each robot joint adds a degree of freedom to the search space.
- **Complex collision constraints:** obstacles, self-collisions, and carried objects restrict valid motions.
- **Narrow passages:** feasible paths may pass through small regions that are difficult to discover.
- **Dynamic feasibility:** paths must admit execution within velocity, acceleration, and actuator limits.
- **Changing environments:** moving obstacles and uncertain geometry require fast updates.

# Challenges in Motion Planning



Find a feasible path through a continuous configuration space.



# Challenges in Control



Execute desired behavior reliably using sensor feedback.

- **Imperfect dynamics models:** friction, payload, and contact forces may be unknown or changing.
- **Noisy and delayed feedback:** measurements provide an imperfect view of the current state.
- **Stability and responsiveness:** aggressive corrections can cause oscillations or instability.
- **Physical limits:** actuator saturation and safety constraints restrict available inputs.
- **Real-time execution:** control inputs must be computed quickly enough to reject disturbances and track the desired motion.

# Challenges in Control



Execute desired behavior reliably using sensor feedback.

**Noisy and delayed feedback:** measurements provide an imperfect view of the current state.



# Challenges in State Estimation



Infer the decision-relevant state from observations and past actions.

- **Partial observability:** occlusions and limited sensor coverage hide relevant information.
- **Noise and ambiguity:** measurements may be inaccurate, and objects may look similar.
- **State aliasing:** different underlying states produce indistinguishable observations or the same estimated representation.
- **Temporal reasoning:** observations must be combined over time while accounting for motion and change.
- **Representing uncertainty:** committing to one estimate can hide alternatives that require different actions.

# Challenges in State Estimation



**State aliasing:** different underlying states produce indistinguishable observations or the same estimated representation.



# Challenges in Multi-Layer Decision Making



Coordinate task planning, motion planning, control, and state estimation.

- **Different representations:** symbolic predicates, geometric configurations, and control states must remain consistent.
- **Cross-layer dependencies:** task choices constrain motion; motion and control limitations determine which task choices are feasible.
- **Different time scales:** high-level planning must coordinate with fast feedback loops.
- **Failure diagnosis and recovery:** decide whether to adjust control, replan motion, revise the task plan, or gather more information.

# Challenges in Multi-Layer Decision Making

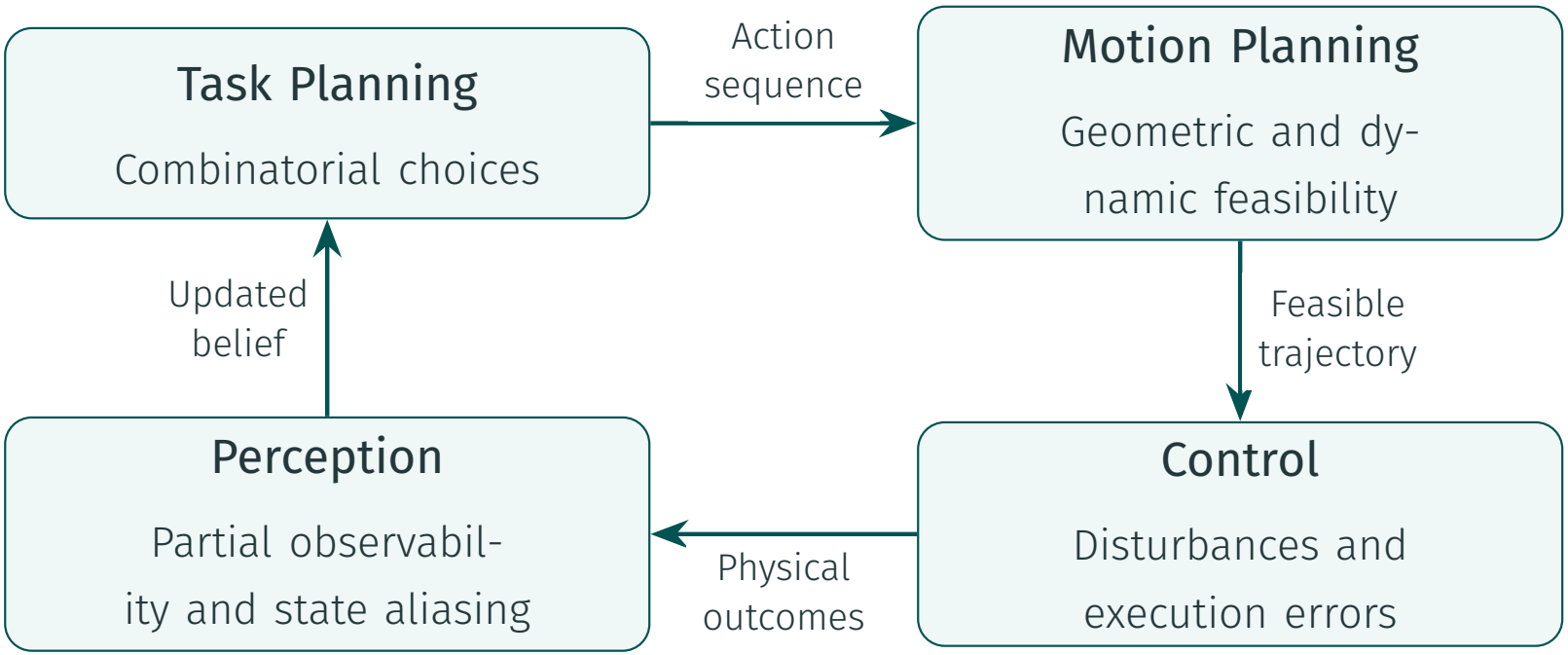


**Failure diagnosis and recovery:** decide whether to adjust control, replan motion, revise the task plan, or gather more information.



# Task and Motion Planning: Key Challenges






Robust behavior requires reasoning, motion, perception, and control to work together.